Octonions, G_2, and Number-Theoretic Obstructions in the Arithmetic of Order Framework: A Topological–Algebraic Explanation for Electroweak Mass under Reinterpretation of the Higgs Mechanism

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Dedication

This work is dedicated to our Parents.
The Arithmetic of Order (AoO) is founded on the natural counting sequence 1 to n to $n + 1$, a principle we learn from our Mothers (XX) and our Fathers (XY).

Abstract

We resolve a sixty-year-old geometric puzzle—the non-existence of a complex structure on the 6-sphere—by revealing its origin in number theory, while simultaneously uncovering a novel mechanism for the origin of electroweak mass. The known geometric obstruction (non-vanishing Nijenhuis tensor) is shown to be a direct consequence of a deeper arithmetic obstruction: the Brauer–Manin obstruction arising from the failure of the equation $x^2 + 1 = 0$ to have solutions over $\mathbb{Q}_p$ for primes $p \equiv 3 \pmod 4$, including the prime $p = 7$. Within the

finitistic Arithmetic of Order (AoO) framework, this unified mathematical "no" becomes a physical "yes": the primes themselves impose intrinsic non-associativity in the octonionic algebra of physical states, which manifests as irreducible geometric torsion on $S^6(\mathbb{R})$. Consequently, the masses of the W and Z bosons are not free parameters but *computable, number-theoretic inevitabilities*, arithmetically mandated by the properties of the primes themselves.

Prime 7 and Limits of the Standard Complex Structure

While standard quantum mechanics employs the complex unit i with $i^2 = -1$ globally, this implicitly assumes a fully integrable complex structure. In the AoO framework, the prime $p = 7$ plays a critical role: the equation $x^2 + 1 = 0$ has no solution in $\mathbb{Q}_7$, the 7-adic completion of the rationals, reflecting a local arithmetic obstruction. This Brauer–Manin obstruction at $p = 7$ prevents a globally consistent complex structure on the rational 6-sphere $S^6(\mathbb{Q}) \subset \mathbb{Q}^7$. Consequently, massless propagation of electroweak gauge bosons, which would require a global i, is forbidden, and the nonzero W and Z masses emerge as a direct, number-theoretically mandated consequence of the primes themselves.

1 Finitistic Foundations and Emergent Geometry

The AoO philosophy posits that reality is fundamentally discrete and computable, built upon rational numbers ($\mathbb{Q}$) generated constructively rather than assumed as a continuum.

- *Rational Octonions via Farey Recursion.* Starting with a finite-yet-unbounded construction of $\mathbb{Q}$ through iterated Farey sequences, the 8-dimensional octonion algebra over $\mathbb{Q}$, denoted $\mathbb{O}_{\mathbb{Q}}$, is defined, along with its 7-dimensional imaginary subspace $\operatorname{Im}\mathbb{O}_{\mathbb{Q}} \cong \mathbb{Q}^7$.

- *The Rational 6-Sphere and its Emergent Continuum.* The unit-norm locus in $\operatorname{Im}\mathbb{O}_{\mathbb{Q}}$ defines the rational 6-sphere, $S^6(\mathbb{Q})$. As denominators in the Farey construction grow, this dense set converges metrically to the real 6-sphere $S^6(\mathbb{R})$, making the continuum an emergent approximation.

- *Coset Geometry: The Rosetta Stone.* The full coset $G_2/(SU(2)_L \times SU(2)_R)$ is an 8-dimensional algebraic variety over $\mathbb{Q}$. A distinguished 6-dimensional submanifold is isomorphic to $S^6(\mathbb{Q})$, realized both as the unit sphere in the imaginary octonions and as the coset $G_2/SU(3)$. This submanifold translates octonionic algebra into physical geometry; the continuum $S^6(\mathbb{R})$ is its emergent, asymptotic limit.

2 Physical Algebra: States, Associators, and Interaction as Geometry

In AoO, fundamental states and their interactions are described by non-associative octonionic algebra, directly inducing the emergent geometry.

- *Physical States.* Fundamental states are tri-octonion vectors:

$$v = (o_1, o_2, o_3) \in (\mathrm{Im}\,\mathbb{O}_{\mathbb{Q}})^3.$$

- *Associator as Internal Tension.* Each state's associator

$$[v] := (o_1 o_2)o_3 - o_1(o_2 o_3)$$

quantifies intrinsic non-associativity, interpreted as "internal tension" or charge.

- *Interaction and Emergent Complex Structure.* Interactions between states v_1, v_2 are governed by the commutator of associators:

$$[[v_1], [v_2]] := [v_1][v_2] - [v_2][v_1] = 2\,[v_2] \times [v_1],$$

where the factor of 2 arises from structure constants. The octonionic cross product defines the almost-complex structure J on $S^6(\mathbb{R})$:

$$J_g(v) = g \times v, \quad g \in S^6.$$

Thus, physical interaction is the geometric structure itself.

3 Geometric Obstruction: Non-Integrability from Non-Associativity

Non-associativity of octonions induces non-integrability of J.

- *Nijenhuis Tensor.* An almost-complex structure J is integrable if and only if

$$N_J(X, Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y] = 0$$

  for all vector fields X, Y.

- *Obstruction Manifest.* For $J_g(v) = g \times v$, $N_J \neq 0$, classically proving that S^6 has no global integrable complex structure.

- *AoO Interpretation.* Non-vanishing N_J is the macroscopic manifestation of irreducible internal tensions $[v] \neq 0$, which directly correspond to geometric torsion.

4 Arithmetic Obstruction: Brauer–Manin and the Prime 7

The "why" of non-integrability lies in number theory.

- *Local-to-Global Principle.* The Hasse principle states that an equation has a solution in $\mathbb{Q}$ only if it has solutions over $\mathbb{R}$ and all $\mathbb{Q}_p$.

- *Quadratic Prototype.* $x^2 + 1 = 0$ is solvable in $\mathbb{Q}_p$ iff $p \equiv 1 \pmod 4$ or $p = 2$. For $p \equiv 3 \pmod 4$ (including $p = 7$), no solution exists; -1 is not a quadratic residue.

- *Brauer–Manin Obstruction.* Extending to the rational octonion algebra B (where $B \otimes_{\mathbb{Q}} \mathbb{R} \cong \mathbb{O}$), the ability to define a complex structure on $S^6(\mathbb{Q}) \subset \mathbb{Q}^7$ depends on whether $B \otimes_{\mathbb{Q}} \mathbb{Q}_p$ splits or remains a division algebra:
  - Over $\mathbb{R}$: B splits; J exists.
  - $p \equiv 1 \pmod 4$: B is expected to split; a local complex structure exists.

- $p \equiv 3 \pmod 4$**, including** $p = 7$: B remains division; **no local complex structure exists.**

The absence of local solutions at $p = 7$ and other $3 \pmod 4$ primes constitutes a Brauer–Manin obstruction, proving that no global integrable complex structure exists on $S^6(\mathbb{Q})$. The prime 7 is thus intrinsic to the obstruction, providing the 7-adic topology $\mathbb{Q}_7$ relevant to the 7-dimensional space $\mathbb{Q}^7$ in which $S^6(\mathbb{Q})$ is defined.

5 Unified Physical Law: Mass as Arithmetic Inevitability

The geometric obstruction ($N_J \neq 0$) is a consequence of the arithmetic obstruction (Brauer–Manin).

- The number-theoretic obstruction **mandates** non-associativity at the fundamental level $[v] \neq 0$.
- Non-associativity **manifests** as irreducible torsion in J on the emergent continuum.
- This torsion acts as unavoidable resistance to propagation; massless electroweak gauge bosons would require a globally integrable complex structure—explicitly forbidden by the primes, including $p = 7$.

Conclusion The W and Z bosons acquire mass not as a contingent Higgs parameter but as a *computable, number-theoretic inevitability*. This reinterprets the Higgs mechanism as a fundamental arithmetic-geometric constraint on the structure of the universe.

Selected References

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